LHCb Detector Acceptance Surfaces

Reusable XY-plane geometries for transport and mask studies

Geometry builder documentation

17 October 2026

Six lower-layout configurations; surfaces and fired-layer labels

Geometry payload and surface meaning

What is stored

Each layout row names a detector surface, its logical layer/station label, its global z in mm, and a Shapely XY geometry. Shapes are reusable pickle files; `detector_setup.pickle` bundles the loaded layers and config.

Build chain

1. Read placements and XY outlines from the ViewLHCb lower ROOT/GDML geometry.
2. Convert outlines to polygons/unions and export named shape pickles.
3. Record each plane's z and logical label in `layout.csv`.
4. Test each transported track crossing against the XY acceptance.

Important distinction

The masks are transverse acceptance surfaces, not a full material model. RICH is represented by projective entry/exit planes; PicoCal by two faces sharing an extracted footprint. The geometry does not model light transport, detailed material interactions, or the field map.

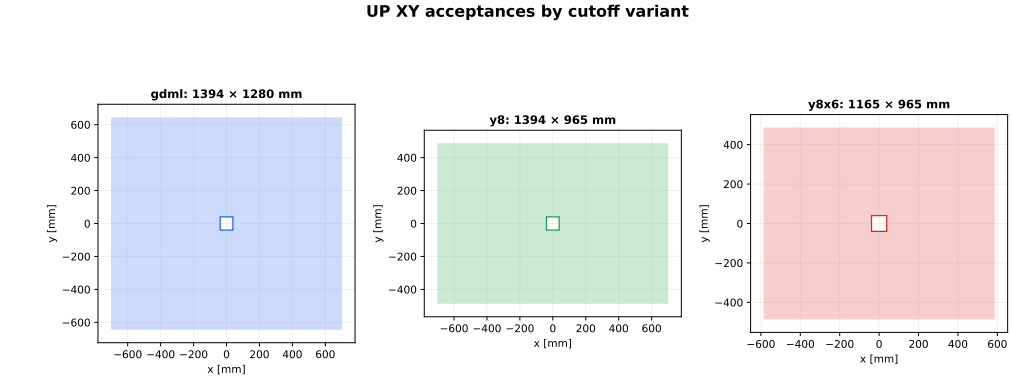
Units are mm. The checked-in layout CSV is authoritative for plane names and exact z values; summary tables round values for readability.

The six available lower configurations

ID	UP border variant	SciFi T1/T2/T3	UP XY envelope [mm]	SciFi XY envelope [mm]
A	GDML nominal	8/8/10	1394 × 1280	5652 × 4799
B	Y8 top/bottom crop	8/8/10	1394 × 965	5652 × 4799
C	Y8X6 top/bottom + sides	8/8/10	1165 × 965	5652 × 4799
D	GDML nominal	10/10/12	1394 × 1280	6707 × 4891
E	Y8 top/bottom crop	10/10/12	1394 × 965	6707 × 4891
F	Y8X6 top/bottom + sides	10/10/12	1165 × 965	6707 × 4891

The six setup IDs are the Cartesian product of UP `gdml`, `y8`, `y8x6` and SciFi `8_8_10`, `10_10_12`. TV, RICH1, MP, RICH2, PicoCal, and Muon are shared across the six choices. The default catalog selection is UP Y8X6 with SciFi 8/8/10.

UP variant XY surfaces and central hole



Variant	Applied border crop	Central hole [mm]
`UP_GDML`	Nominal ViewLHCb/GDML borders	77.3 × 79.9
`UP_Y8`	Remove 8 chips at top and bottom	77.3 × 79.9
`UP_Y8X6`	Remove 8 top/bottom and 6 left/right chips	77.3 × 79.9

How the acceptance outlines are derived

Subsystem	Stored planes	Construction used in these setups
TV / VELO	50	Pair nearby sensor-face groups within 2 mm in z; union each pair in XY. This yields one fired layer per TV layer.
RICH1/2	2 each	Project angular boundaries from the origin using $x = z\tan\theta_x$, $y = z\tan\theta_y$ at entry and exit z. Simplified planes only.
UP	16	Keep the extracted surfaces and group four consecutive surfaces per logical UP layer. Apply the selected border variant to XY.
MP	5	Union the extracted tile outlines into five logical XY layer surfaces, avoiding hundreds of per-tile planes.
SciFi / FT	12	Construct four X/U/V/X acceptance shapes at each T station; use module-count width, fixed height, central aperture, and stereo rotations.
PicoCal	2	Use the extracted front-region footprint as entry and exit faces spanning its z extent.
Muon	4	Build a centered outer rectangle minus a centered rectangular beam aperture at M2–M5.

The generated surface views on the following slides are from the selected setup's geometry summary. They show XY acceptance and Z placement; they are not to scale across detector systems.

LHCb XY surface setup: lhcb_lower_up_gdml_ft_8_8_10

Selected reusable detector planes for transport and acceptance masks.

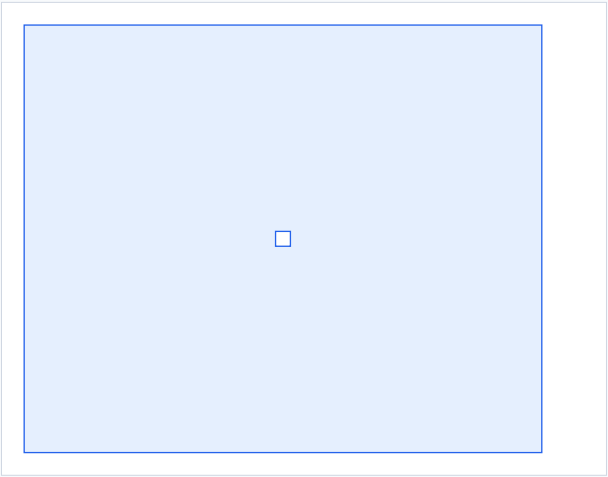
detector	surfaces	fired layers	z range [mm]	XY envelope [mm]	description
TV	50	50	-287.5 to 557.6	90.3 x 92.0	VELO/TV layers formed by unioning the nearby-z sensor faces in each layer.
RICH1	2	2	795.0 to 2237.2	1384.1 x 1142.5	Projective RICH1 radiator entry/exit surfaces from angular acceptance.
UP	16	4	2314.0 to 2656.0	1393.7 x 1279.8	Upstream Tracker surfaces; choose nominal GDML or Yisheng-inspired border crops.
MP	5	5	7665.5 to 9132.5	3288.4 x 672.6	Five Mighty Pixel layers, each formed from the union of its extracted tiles.
FT	12	12	7902.7 to 9384.7	5652.3 x 4798.7	Mighty Tracker X,U,V,X stations; choose 8/10/12 module widths per T station.
RICH2	2	2	9487.5 to 11839.2	2855.1 x 2375.8	Projective RICH2 radiator entry/exit surfaces from angular acceptance.
PicoCal	2	2	12565.9 to 12601.0	7801.6 x 6338.8	Two front-region PicoCal surfaces, entry and exit, as a simple volume proxy.
MUON	4	4	15270.0 to 18870.0	12119.2 x 10057.3	Muon stations M2-M5, rectangular acceptances with central beam holes.

Configuration

{"ft": {"stations": {"T1": 8, "T2": 8, "T3": 10}}, "include": ["TV", "RICH1", "UP", "MP", "FT", "RICH2", "PicoCal", "MUON"], "name": "lhcb_lower_up_gdml_ft_8_8_10", "source": "ViewLHCb lower ROOT/GDML extracted surfaces", "tv": {"grouping": "paired_

RICH1 XY surfaces

2 planes, z = 795.000 to 2237.200 mm, XY envelope = 1384.1 x 1142.5 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

795.0 mm — 2237.2 mm

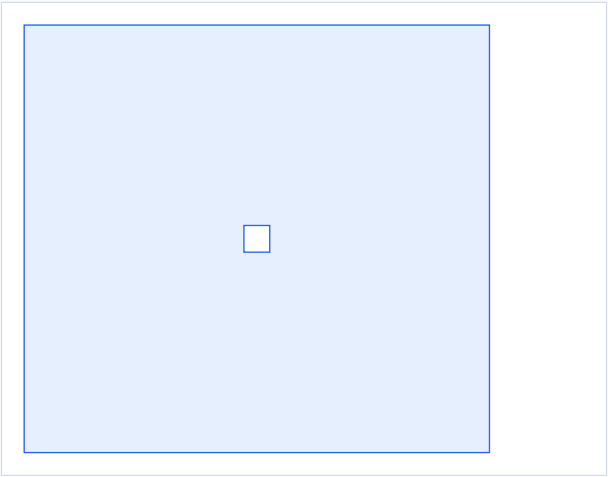
Layers

rich1_layerEntry: z=795.000

rich1_layerExit: z=2237.200

UP XY surfaces

16 planes, z = 2314.031 to 2655.969 mm, XY envelope = 1393.7 x 1279.8 mm



x-y view; holes are shown by white cutouts where present

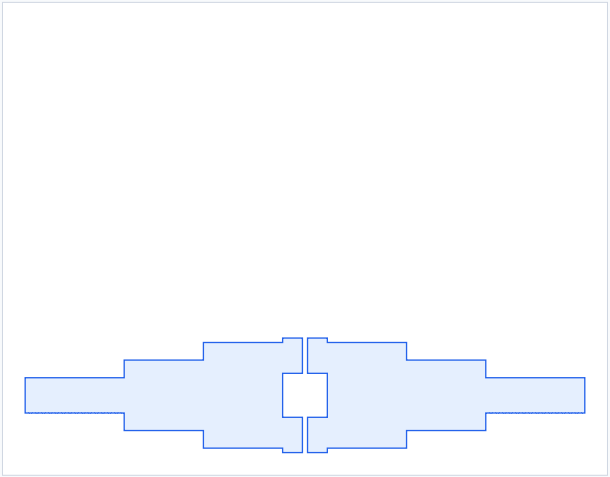


Layers

UP_layer1_surface1: z=2314.031
UP_layer1_surface2: z=2314.232
UP_layer1_surface3: z=2320.768
UP_layer1_surface4: z=2320.969
UP_layer2_surface1: z=2369.031
UP_layer2_surface2: z=2369.232
UP_layer2_surface3: z=2375.768
UP_layer2_surface4: z=2375.969
UP_layer3_surface1: z=2594.031
UP_layer3_surface2: z=2594.232
UP_layer3_surface3: z=2600.768
UP_layer3_surface4: z=2600.969
UP_layer4_surface1: z=2649.031
UP_layer4_surface2: z=2649.232
UP_layer4_surface3: z=2655.768
UP_layer4_surface4: z=2655.969

MP XY surfaces

5 planes, z = 7665.500 to 9132.500 mm, XY envelope = 3288.4 x 672.6 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

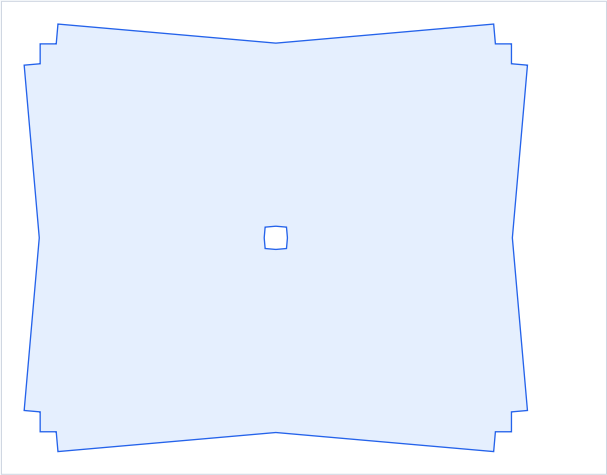


Layers

MP_layer1: z=7665.500
MP_layer2: z=7825.500
MP_layer3: z=8380.500
MP_layer4: z=8972.500
MP_layer5: z=9132.500

FT XY surfaces

12 planes, z = 7902.700 to 9384.700 mm, XY envelope = 5652.3 x 4798.7 mm



x-y view; holes are shown by white cutouts where present

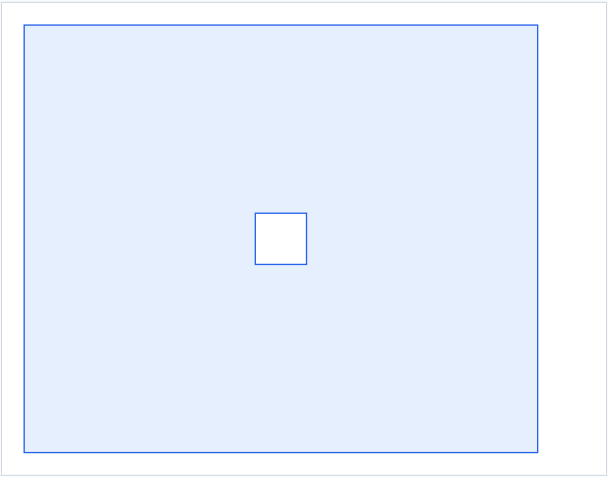


Layers

scifi_layer1_T1_X1: z=7902.700
scifi_layer2_T1_U: z=7947.699
scifi_layer3_T1_V: z=8032.701
scifi_layer4_T1_X2: z=8077.700
scifi_layer5_T2_X1: z=8524.700
scifi_layer6_T2_U: z=8569.699
scifi_layer7_T2_V: z=8654.701
scifi_layer8_T2_X2: z=8699.700
scifi_layer9_T3_X1: z=9209.700
scifi_layer10_T3_U: z=9254.699
scifi_layer11_T3_V: z=9339.701
scifi_layer12_T3_X2: z=9384.700

RICH2 XY surfaces

2 planes, z = 9487.500 to 11839.200 mm, XY envelope = 2855.1 x 2375.8 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

9487.5 mm — 11839.2 mm

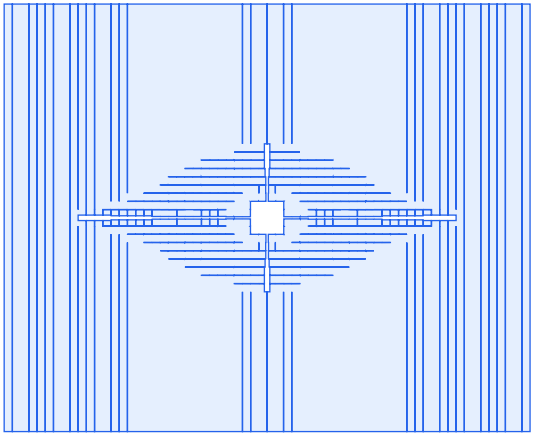
Layers

rich2_layerEntry: z=9487.500

rich2_layerExit: z=11839.200

PicoCal XY surfaces

2 planes, z = 12565.869 to 12601.012 mm, XY envelope = 7801.6 x 6338.8 mm



x-y view; holes are shown by white cutouts where present

z placement of all planes

12565.9 mm — 12601.0 mm

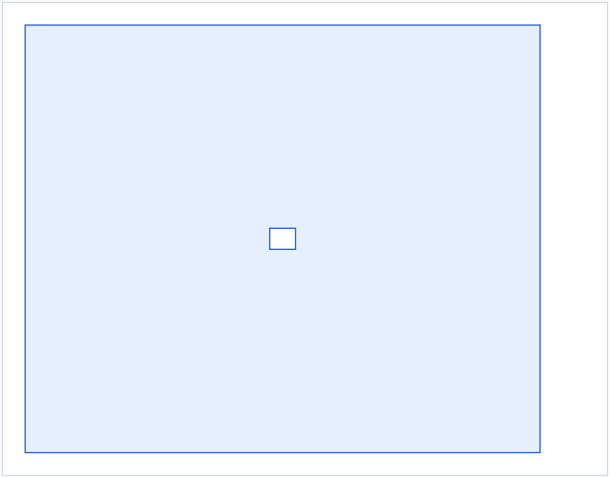
Layers

picocal_layerEntry: z=12565.869

picocal_layerExit: z=12601.012

MUON XY surfaces

4 planes, z = 15270.000 to 18870.000 mm, XY envelope = 12119.2 x 10057.3 mm



x-y view; holes are shown by white cutouts where present



Layers

muon_m2: z=15270.000
muon_m3: z=16470.000
muon_m4: z=17670.000
muon_m5: z=18870.000

Available UP and FT XY variants

Use this page to compare choices before building the final setup.

gdml



1393.7 x 1279.8 mm

y8



1393.7 x 964.9 mm

y8x6



1164.7 x 964.9 mm

FT 8 X



4234.4 x 4354.0 mm

FT 8 U



4597.8 x 4706.5 mm

FT 8 V



4597.8 x 4706.5 mm

FT 10 X



5293.0 x 4354.0 mm

FT 10 U



5652.3 x 4798.7 mm

FT 10 V



5652.3 x 4798.7 mm

FT 12 X



6351.6 x 4354.0 mm

FT 12 U



6706.9 x 4891.0 mm

FT 12 V



6706.9 x 4891.0 mm

SciFi module counts and stereo convention

Surface dimensions before rotation

Module width: 529.3 mm
Active height: 4354 mm
Central square opening: 260 × 260 mm

$W_x = 529.3N$ mm, for $N \in \{8, 10, 12\}$

T1 and T2 use the first module count in the selected triplet; T3 uses the third. For example, 8/8/10 means widths 4234.4, 4234.4, and 5293.0 mm.

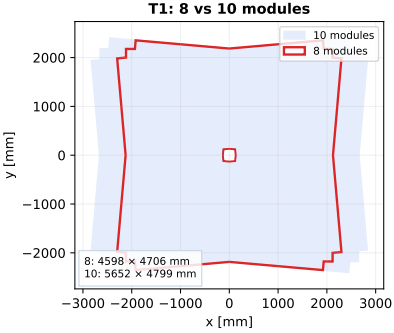
Four surfaces per station

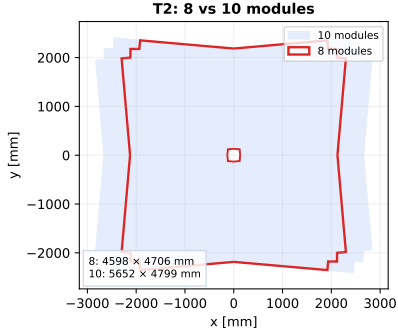
Layer	T1 z	T2 z	T3 z [mm]
X1	7902.700	8524.700	9209.700
U	7947.699	8569.699	9254.699
V	8032.701	8654.701	9339.701
X2	8077.700	8699.700	9384.700

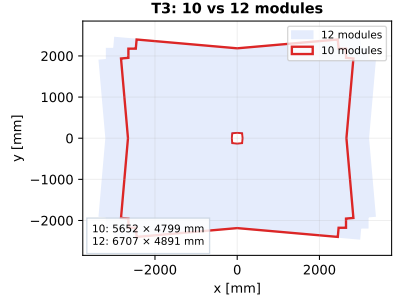
Rotations: X1/X2 0°, U −5°, V +5°.

The exact z coordinates are retained from extracted baseline X/U/V/X groups across all module-count choices. The layout CSV is authoritative.

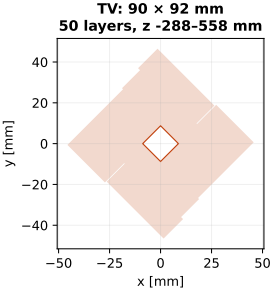
SciFi station XY envelopes

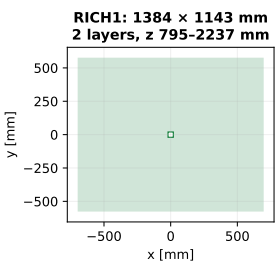


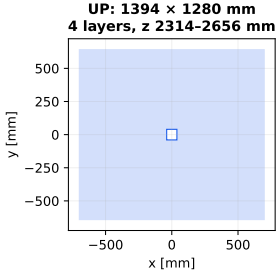


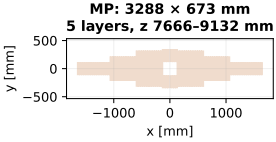


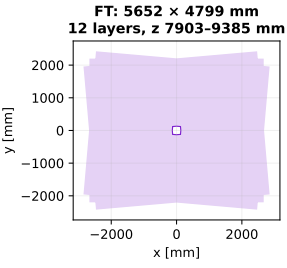
Full detector XY surfaces: reference configuration A

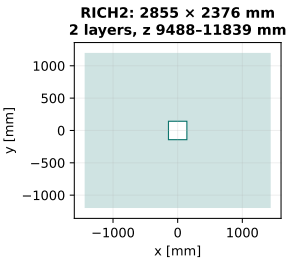


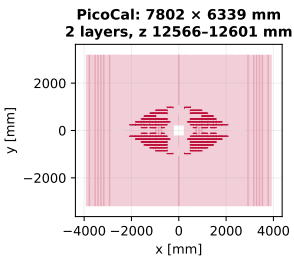


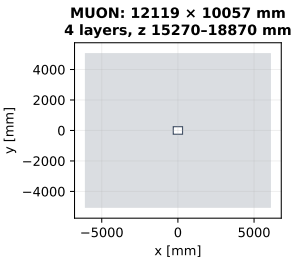












Other than UP and SciFi, these XY surfaces are shared by all six configurations.

Logical fired-layer labels

Output	Range	Counting rule
`nTV`	0–50	Number of paired-face TV layer unions crossed inside acceptance.
`nUP`	0–4	A logical layer fires if the crossing is inside the union of its four grouped source surfaces.
`nMP`	0–5	One hit per unioned MP layer.
`nFT`	0–12	One hit per X/U/V/X surface at T1, T2, and T3.
`nMuon`	0–4	One hit per M2, M3, M4, or M5 station.
`hasRich1Entry/Exit`	Boolean	Crossing inside the corresponding RICH1 face. Same convention for RICH2.
`hasRichEntry/Exit`	Boolean	OR of the corresponding entry/exit hit in RICH1 and RICH2.
`hasPicoCal`	Boolean	At least one of the two PicoCal faces is hit.

The loader can append a configuration label to output columns, for example `nFT_Run5`, so multiple setups can be evaluated on the same track. Plane crossings are supplied by the transport/projector; this package tests the crossing point against the selected XY shape and counts the fired labels.

Provenance and files to use

- `detector_setups/catalog.json`: available choices and default setup.
- `detector_setups/<setup>/config.json`: resolved setup options.
- `detector_setups/<setup>/layout.csv`: exact plane names, labels, and global z values.
- `detector_setups/shared_shapes/` and `up_variants/`: reusable source shapes for the six catalog entries.
- `detector_setups/<setup>/detector_setup.pickle`: loaded geometry payload used by the mask loader.

Scope note. These are the current reusable acceptance configurations. They are designed for geometry acceptance comparisons, not detector simulation. The detailed plane definitions and saved XY shapes take precedence over the rounded envelope dimensions shown in this deck.